Lab_Assignment_14.3

AI Assisted Coding B37 2303A52077

Task 1: AI-Assisted Personal Portfolio Website

Scenario

A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure with sections:
    - Home
    - About
    - Projects
    - Contact
2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.
3. Customize AI-generated CSS to add:
    - Hover effects on project cards
    - Smooth scrolling navigation

Expected Outcome

- Clean and well-structured portfolio website
- Responsive layout for desktop and mobile screens

- Interactive project cards

Prompt:

Create a **personal portfolio website** using semantic HTML with sections for **Home, About, Projects, and Contact**. Use AI assistance to generate **responsive CSS with Flexbox or Grid**, and enhance the design with **hover effects on project cards and smooth scrolling navigation**. Ensure the layout works well on both desktop and mobile devices.

Code:

```
import webbrowser

import os


html_content = """<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>My Portfolio</title>

    <style>

        body {

            font-family: Arial, sans-serif;
```

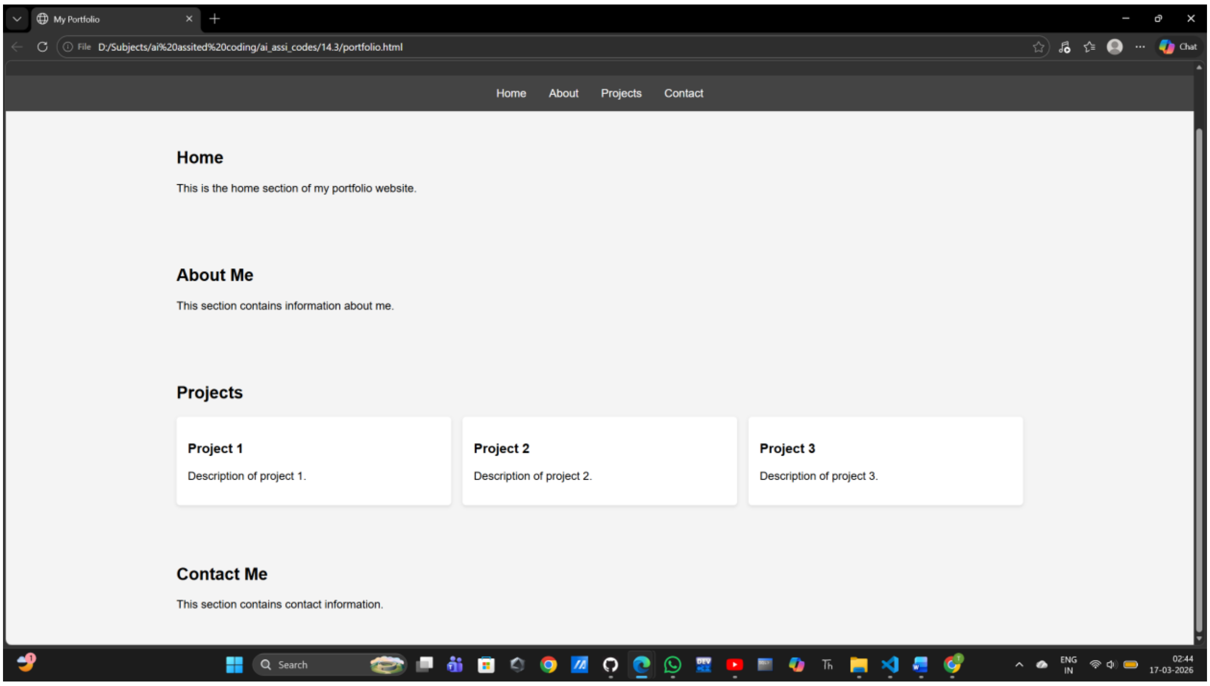
```
    margin: 0;
    padding: 0;
    background-color: #f4f4f4;
}
header {
    background-color: #333;
    color: white;
    padding: 1em 0;
    text-align: center;
}
nav {
    display: flex;
    justify-content: center;
    background-color: #444;
}
nav a {
    color: white;
    padding: 1em;
    text-decoration: none;
}
nav a:hover {
    background-color: #555;
}
```

```
section {
    padding: 2em;
    max-width: 1200px;
    margin: auto;
}
.projects {
    display: grid;
    grid-template-columns: repeat(auto-fit,
minmax(300px, 1fr));
    gap: 1em;
}
.project-card {
    background-color: white;
    padding: 1em;
    border-radius: 5px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.1);
    transition: transform 0.3s ease, box-shadow 0.3s ease;
}
.project-card:hover {
    transform: translateY(-10px);
    box-shadow: 0 4px 10px rgba(0,0,0,0.2);
}
@media (max-width: 600px) {
```

```
            nav {
                flex-direction: column;
                align-items: center;
            }
        }
    </style>
</head>
<body>
    <header>
        <h1>Welcome to My Portfolio</h1>
    </header>
    <nav>
        <a href="#home">Home</a>
        <a href="#about">About</a>
        <a href="#projects">Projects</a>
        <a href="#contact">Contact</a>
    </nav>
    <section id="home">
        <h2>Home</h2>
        <p>This is the home section of my portfolio
website.</p>
    </section>
    <section id="about">
```

```
    <h2>About Me</h2>
    <p>This section contains information about me.</p>
</section>
<section id="projects">
    <h2>Projects</h2>
    <div class="projects">
        <div class="project-card">
            <h3>Project 1</h3>
            <p>Description of project 1.</p>
        </div>
        <div class="project-card">
            <h3>Project 2</h3>
            <p>Description of project 2.</p>
        </div>
        <div class="project-card">
            <h3>Project 3</h3>
            <p>Description of project 3.</p>
        </div>
    </div>
</section>
<section id="contact">
    <h2>Contact Me</h2>
    <p>This section contains contact information.</p>
```

```
    </section>

    <script>
        // Smooth scrolling for navigation links
        document.querySelectorAll('nav a').forEach(anchor => {
            anchor.addEventListener('click', function(e) {
                e.preventDefault();
                document.querySelector(this.getAttribute('href')).scr
ollIntoView({
                    behavior: 'smooth'
                });
            });
        });
    </script>
</body>
</html>"""

# Write HTML to file and open in browser
html_file = "portfolio.html"
with open(html_file, "w") as f:
    f.write(html_content)

webbrowser.open("file://" + os.path.realpath(html_file))
```



Explanation:

AI tools assisted in generating the HTML structure by providing a clear and organized layout for the portfolio website, including semantic sections for Home, About, Projects, and Contact. The AI also helped in creating responsive CSS using Flexbox and Grid, ensuring that the design adapts well to different screen sizes. Additionally, AI suggested enhancements such as hover effects on project cards to improve user interaction and smooth scrolling navigation for a better user experience when navigating through the sections of the website. These improvements contribute to a more engaging and accessible portfolio site across various devices.

Task 2: AI-Generated Restaurant Landing Page

Scenario

A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.

Tasks

1. Use Copilot to create a navigation bar with links:
    - Home
    - Menu
    - Gallery
    - Contact
2. Generate a menu section styled with CSS cards.
3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.
4. Customize animations or transitions for better user experience.

Expected Outcome

- Fully functional restaurant landing page
- Responsive navigation bar
- Working image slider with smooth transitions

Prompt:

#Create a restaurant landing page using HTML, CSS, and JavaScript with sections for Home, Menu, Gallery, and Contact.

Use AI assistance to build a responsive navigation bar, design the menu section with CSS cards, and implement a JavaScript

image slider for the gallery. Enhance the page with animations or smooth transitions for better user experience.

Code:

```
import webbrowser
import os
html_content = """<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Landing Page</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f8f8f8;
        }
```

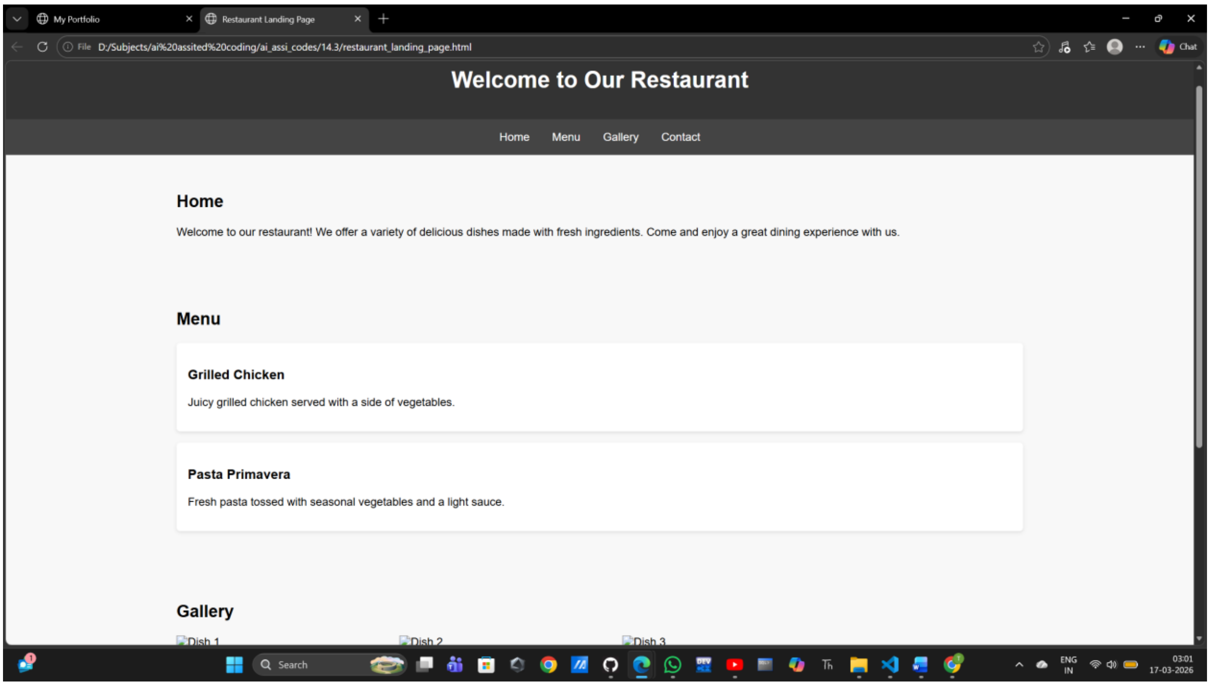
```
header {
    background-color: #333;
    color: white;
    padding: 1em 0;
    text-align: center;
}
nav {
    display: flex;
    justify-content: center;
    background-color: #444;
}
nav a {
    color: white;
    padding: 1em;
    text-decoration: none;
}
nav a:hover {
    background-color: #555;
}
section {
    padding: 2em;
    max-width: 1200px;
    margin: auto;
```

```
}
.menu-card {
    background-color: white;
    padding: 1em;
    border-radius: 5px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.1);
    margin-bottom: 1em;
}
.gallery {
    display: flex;
    overflow-x: auto;
}
.gallery img {
    width: 300px;
    height: 200px;
    object-fit: cover;
    margin-right: 1em;
}
@media (max-width: 600px) {
    .gallery {
        flex-direction: column;
    }
    .gallery img {
```

```
                width: 100%;
                height: auto;
                margin-bottom: 1em;
            }
        }
    </style>
</head>
<body>
    <header>
        <h1>Welcome to Our Restaurant</h1>
    </header>
    <nav>
        <a href="#home">Home</a>
        <a href="#menu">Menu</a>
        <a href="#gallery">Gallery</a>
        <a href="#contact">Contact</a>
    </nav>
    <section id="home">
        <h2>Home</h2>
        <p>Welcome to our restaurant! We offer a variety of delicious dishes made with fresh ingredients. Come and enjoy a great dining experience with us.</p>
    </section>
```

```
    <section id="menu">
        <h2>Menu</h2>
        <div class="menu-card">
            <h3>Grilled Chicken</h3>
            <p>Juicy grilled chicken served with a side of
vegetables.</p>
        </div>
        <div class="menu-card">
            <h3>Pasta Primavera</h3>
            <p>Fresh pasta tossed with seasonal vegetables and a
light sauce.</p>
        </div>
    </section>
    <section id="gallery">
        <h2>Gallery</h2>
        <div class="gallery">
            <img src="https://via.placeholder.com/300x200"
alt="Dish 1">
            <img src="https://via.placeholder.com/300x200"
alt="Dish 2">
            <img src="https://via.placeholder.com/300x200"
alt="Dish 3">
        </div>
    </section>
```

```
    <section id="contact">
        <h2>Contact</h2>
        <p>Email: info@restaurant.com</p>
        <p>Phone: (123) 456-7890</p>
    </section>
</body>
</html>"""
html_file = "restaurant_landing_page.html"
with open(html_file, "w") as file:
    file.write(html_content)
```



Explanation:

AI tools were instrumental in generating the navigation bar, menu card layout, and gallery slider for the restaurant landing page. The AI assisted in creating a responsive navigation bar that adapts to different screen sizes, ensuring a seamless user experience on both desktop and mobile devices. For the menu card layout, AI helped design visually appealing cards that effectively showcase the dishes offered by the restaurant, making it easy for users to browse through the menu. The gallery slider was also created with AI assistance, allowing for smooth transitions between images and enhancing the visual appeal of the gallery section. Animations and transitions play a crucial role in improving the user experience by providing visual feedback and making interactions more engaging. For instance, hover effects on the navigation links and menu cards create a dynamic feel, encouraging users to explore the content. Smooth transitions in the gallery slider enhance the overall aesthetic and make it easier for users to navigate through images without abrupt changes, contributing to a more polished and professional landing page.

Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:
    - Name
    - Email

   - Phone Number
   - Department
   - Event Selection
2. Use AI assistance to style the form using CSS for readability and accessibility.
3. Implement JavaScript validation using Copilot:
   - Email format validation
   - Phone number length validation
   - Required field checks

Expected Outcome

- User-friendly and visually appealing form
- Real-time validation messages
- Prevention of invalid submissions

Prompt:

Create an **event registration form** using HTML with fields for **Name, Email, Phone Number, Department, and Event Selection**. Use AI assistance to style the form with **CSS for readability and accessibility**, and implement **JavaScript validation** for email format, phone number length, and required fields.

Code:

#Create an event registration form using HTML with fields for Name, Email, Phone Number, Department, and Event Selection.

Use AI assistance to style the form with CSS for readability and accessibility, and implement JavaScript validation for

email format, phone number length, and required fields.

```
import webbrowser
import os
html_content = """<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Event Registration Form</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f8f8f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
```

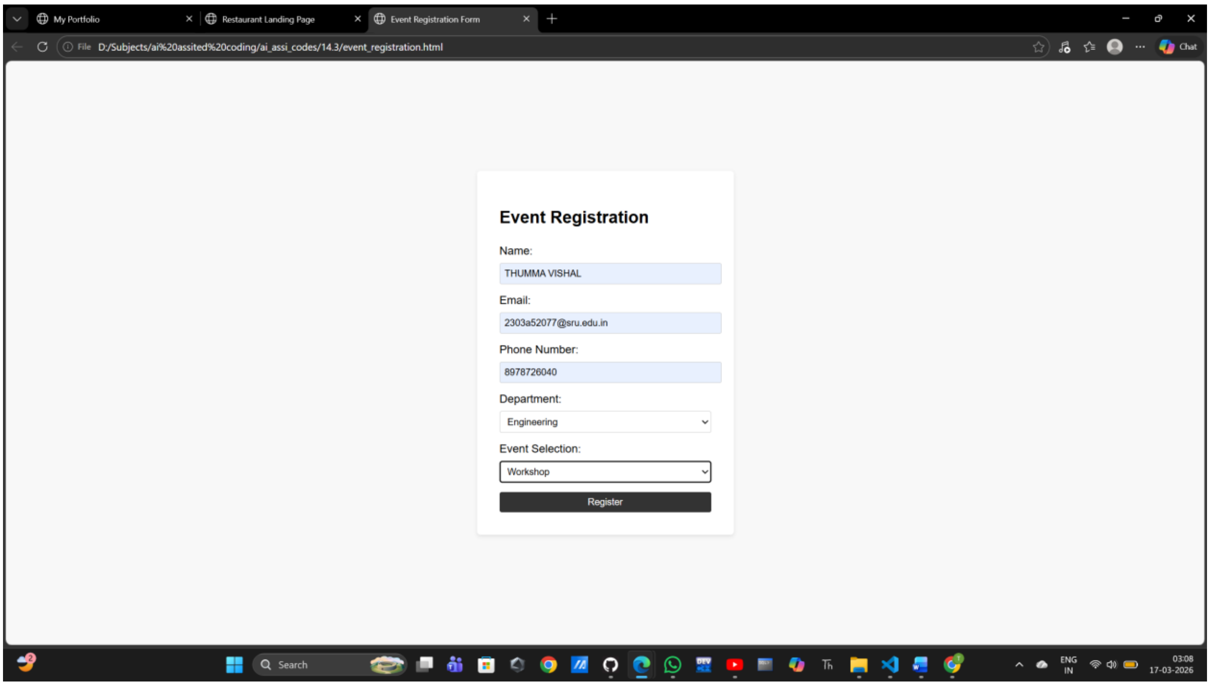
```
    margin: 0;
}
.registration-form {
    background-color: white;
    padding: 2em;
    border-radius: 5px;
    box-shadow: 0 2px 5px rgba(0,0,0,0.1);
    width: 300px;
}
.registration-form h2 {
    margin-bottom: 1em;
}
.registration-form label {
    display: block;
    margin-bottom: 0.5em;
}
.registration-form input, .registration-form select {
    width: 100%;
    padding: 0.5em;
    margin-bottom: 1em;
    border: 1px solid #ccc;
    border-radius: 3px;
}
```

```
        .registration-form button {
            width: 100%;
            padding: 0.5em;
            background-color: #333;
            color: white;
            border: none;
            border-radius: 3px;
            cursor: pointer;
        }
        .registration-form button:hover {
            background-color: #555;
        }
    </style>
</head>
<body>
    <form class="registration-form" id="eventForm">
        <h2>Event Registration</h2>
        <label for="name">Name:</label>
        <input type="text" id="name" name="name" required>

        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>
```

```
        <label for="phone">Phone Number:</label>
        <input type="tel" id="phone" name="phone" required
pattern="[0-9]{10}">

        <label for="department">Department:</label>
        <select id="department" name="department" required>
            <option value="">Select Department</option>
            <option value="sales">Sales</option>
            <option value="marketing">Marketing</option>
            <option value="engineering">Engineering</option>
            <option value="hr">Human Resources</option>
        </select>

        <label for="event">Event Selection:</label>
        <select id="event" name="event" required>
            <option value="">Select Event</option>
            <option value="workshop">Workshop</option>
            <option value="seminar">Seminar</option>
            <option value="networking">Networking
Event</option>
        </select>

        <button type="submit">Register</button>
```

```
    </form>

    <script>
        document.getElementById('eventForm').addEventListen
er('submit', function(event) {
            event.preventDefault();

            const email =
document.getElementById('email').value;
            const phone =
document.getElementById('phone').value;

            const emailPattern =
/^[^\\s@]+@[^\\s@]+\\.[^\\s@]+$/;
            if (!emailPattern.test(email)) {
                alert('Please enter a valid email address.');
                return;
            }

            if (phone.length !== 10 || isNaN(phone)) {
                alert('Please enter a valid 10-digit phone number.');
                return;
            }
```

```
            alert('Registration successful!');
            this.reset();
        });
    </script>
</body>
</html>"""

with open('event_registration.html', 'w') as f:
    f.write(html_content)

webbrowser.open('file://' +
os.path.realpath('event_registration.html'))
```



Explanation:

AI generate the HTML form structure by providing a template that included all necessary form elements with appropriate attributes. The CSS styling was generated to create a clean, modern look with responsive design principles. The JavaScript validation logic was created to ensure data integrity by checking email format and phone number length before submission. Real-time validation improves data accuracy by providing immediate feedback to users about input errors, and enhances user experience by reducing the likelihood of submission failures due to invalid data. This allows users to correct their input on the spot, leading to a smoother registration process and higher quality data for event organizers.

Task 4: AI-Assisted E-Commerce Product Page

Scenario

A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks

1. Use Copilot to generate a grid-based product catalog using HTML and CSS.
2. Implement an “Add to Cart” feature using JavaScript.
3. Modify AI-generated code to include:
    - Cart item counter at the top-right corner
    - Visual feedback when a product is added

Expected Outcome

- Functional product catalog
- Working cart counter
- Interactive user experience

Prompt:

#Create a **basic e-commerce product page** using HTML, CSS, and JavaScript. Use AI assistance to generate a **grid-based product catalog**, implement an **“Add to Cart” feature**, and update the interface to show a **cart item counter at the top-right corner**. Add **visual feedback when a product is added to the cart** to improve user interaction.

Code:

```
#Create a basic e-commerce product page using HTML, CSS, and JavaScript.

# Use AI assistance to generate a grid-based product catalog, implement an “Add to Cart” feature,

# and update the interface to show a cart item counter at the top-right corner.

# Add visual feedback when a product is added to the cart to improve user interaction.


import webbrowser

import os

html_content = """<!DOCTYPE html>

<html lang="en">
```

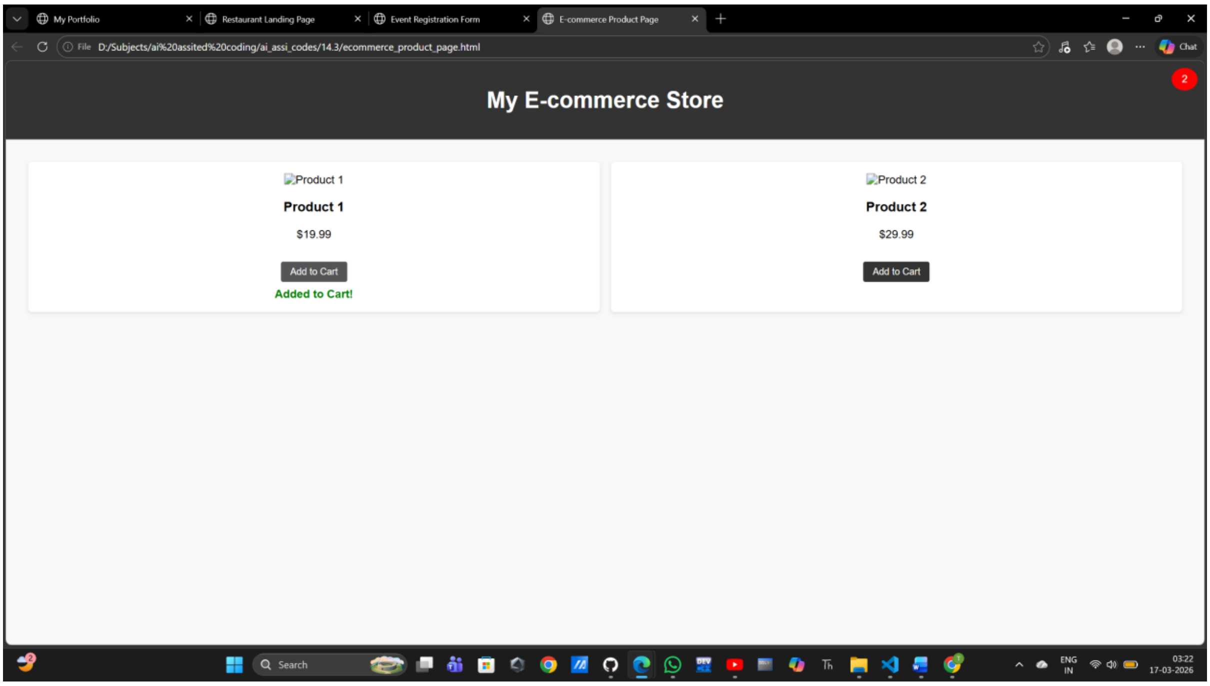
```
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>E-commerce Product Page</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f8f8f8;
        }
        header {
            background-color: #333;
            color: white;
            padding: 1em 0;
            text-align: center;
            position: relative;
        }
        .cart-counter {
            position: absolute;
            top: 10px;
            right: 10px;
```

```
      background-color: red;
      color: white;
      border-radius: 50%;
      padding: 0.5em 1em;
      font-size: 0.9em;
    }
    .product-grid {
      display: grid;
      grid-template-columns: repeat(auto-fit,
minmax(200px, 1fr));
      gap: 1em;
      padding: 2em;
    }
    .product-card {
      background-color: white;
      padding: 1em;
      border-radius: 5px;
      box-shadow: 0 2px 5px rgba(0,0,0,0.1);
      text-align: center;
    }
    .product-card img {
      width: 100%;
      height: auto;
```

```
        }
        .add-to-cart {
            margin-top: 1em;
            padding: 0.5em 1em;
            background-color: #333;
            color: white;
            border: none;
            border-radius: 3px;
            cursor: pointer;
        }
        .add-to-cart:hover {
            background-color: #555;
        }
        .added-feedback {
            display: none;
            margin-top: 0.5em;
            color: green;
            font-weight: bold;
        }
    </style>
</head>
<body>
    <header>
```

```
        <h1>My E-commerce Store</h1>
        <div class="cart-counter" id="cartCounter">0</div>
    </header>
    <div class="product-grid">
        <div class="product-card">
            <img src="https://via.placeholder.com/200"
alt="Product 1">
            <h3>Product 1</h3>
            <p>$19.99</p>
            <button class="add-to-cart"
onclick="addToCart()">Add to Cart</button>
            <div class="added-feedback">Added to Cart!</div>
        </div>
        <div class="product-card">
            <img src="https://via.placeholder.com/200"
alt="Product 2">
            <h3>Product 2</h3>
            <p>$29.99</p>
            <button class="add-to-cart"
onclick="addToCart()">Add to Cart</button>
            <div class="added-feedback">Added to Cart!</div>
        </div>
        <!-- More product cards can be added here -->
    </div>
```

```
    <script>
        let cartCount = 0;
        function addToCart() {
            cartCount++;
            document.getElementById('cartCounter').innerText =
cartCount;
            const feedback = event.target.nextElementSibling;
            feedback.style.display = 'block';
            setTimeout(() => {
                feedback.style.display = 'none';
            }, 2000);
        }
    </script>
</body>
</html>"""
html_file = "ecommerce_product_page.html"
with open(html_file, "w") as file:
    file.write(html_content)
```



Explanation:

AI tools were instrumental in generating the product catalog layout, cart functionality, and UI interactions for the e-commerce product page. The AI assisted in creating a responsive grid-based layout for the product catalog, ensuring that the products are displayed in an organized and visually appealing manner across different screen sizes. For the cart functionality, AI helped implement the "Add to Cart" feature, allowing users to easily add products to their cart and see the updated cart item counter in real-time. The visual feedback feature was also designed with AI assistance, providing users with immediate confirmation when a product is added to the cart, which enhances the overall user experience by making interactions more engaging and satisfying. The cart counter at the top-right corner serves as a constant reminder of the items in the user's cart, encouraging them to continue shopping or proceed to checkout, while the visual feedback adds a layer of interactivity that keeps users informed and engaged throughout their shopping experience.

Task 5: Create a Responsive Web Page Layout

Instructions:

- Design a basic web page layout with a **header, main content area, and footer** using HTML and CSS.
- Use AI to assist in generating **responsive CSS** for different screen sizes.
- Ensure the layout is clean and visually organized.

Expected Output:

- A responsive web page with:
  - Header with navigation links
  - Main section with placeholder text/images
  - Footer with copyright or contact info
- Layout adapts correctly to desktop, tablet, and mobile screen sizes.

Prompt:

#Create a responsive web page layout using HTML and CSS that includes a header with navigation links,

a main content section with placeholder text or images, and a footer with copyright or contact information.

Use AI assistance to generate responsive CSS for desktop, tablet, and mobile screens.

Ensure the layout remains clean and visually organized across all devices.

Code:

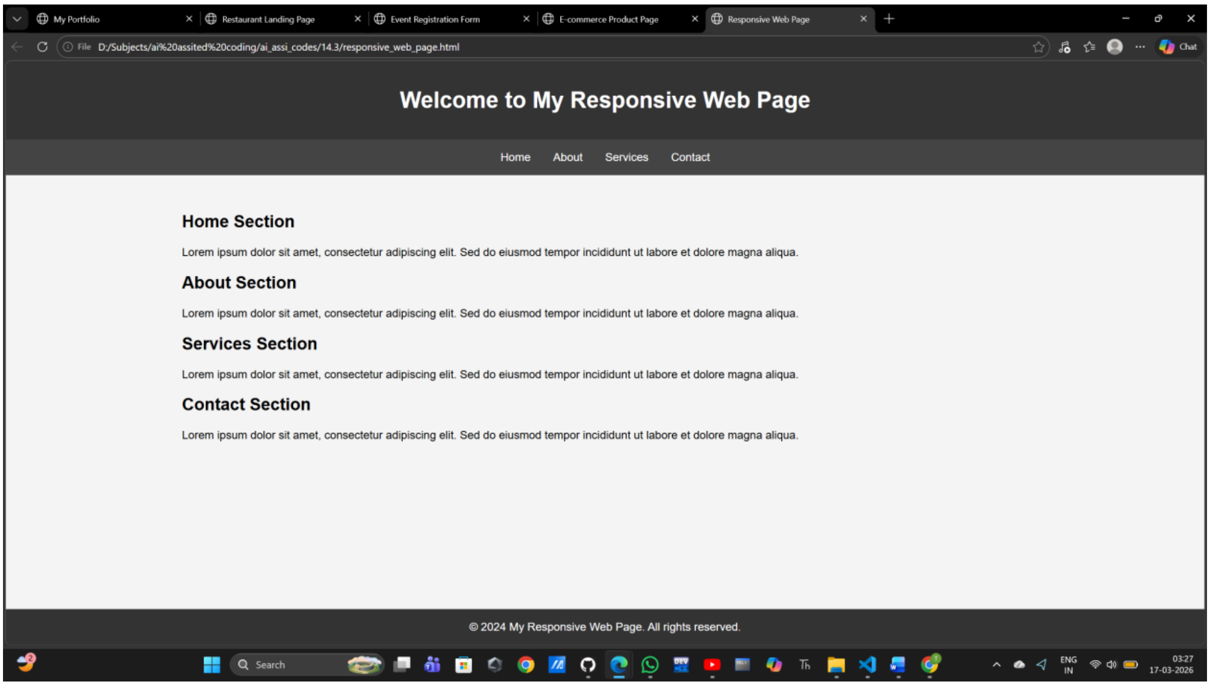
```
import webbrowser
import os
html_content = """<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Responsive Web Page</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f4f4f4;
        }
        header {
            background-color: #333;
```

```css
    color: white;
    padding: 1em 0;
    text-align: center;
}
nav {
    display: flex;
    justify-content: center;
    background-color: #444;
}
nav a {
    color: white;
    padding: 1em;
    text-decoration: none;
}
nav a:hover {
    background-color: #555;
}
main {
    padding: 2em;
    max-width: 1200px;
    margin: auto;
}
footer {
```

```
            background-color: #333;
            color: white;
            text-align: center;
            padding: 1em 0;
            position: fixed;
            width: 100%;
            bottom: 0;
        }
        @media (max-width: 768px) {
            nav {
                flex-direction: column;
                align-items: center;
            }
        }
        @media (max-width: 480px) {
            main {
                padding: 1em;
            }
        }
    </style>
</head>
<body>
    <header>
```

```
        <h1>Welcome to My Responsive Web Page</h1>
    </header>
    <nav>
        <a href="#home">Home</a>
        <a href="#about">About</a>
        <a href="#services">Services</a>
        <a href="#contact">Contact</a>
    </nav>
    <main>
        <h2>Home Section</h2>
        <p>Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Sed do eiusmod tempor incididunt ut labore et dolore
magna aliqua.</p>
        <h2>About Section</h2>
        <p>Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Sed do eiusmod tempor incididunt ut labore et dolore
magna aliqua.</p>
        <h2>Services Section</h2>
        <p>Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Sed do eiusmod tempor incididunt ut labore et dolore
magna aliqua.</p>
        <h2>Contact Section</h2>
        <p>Lorem ipsum dolor sit amet, consectetur adipiscing
elit. Sed do eiusmod tempor incididunt ut labore et dolore
magna aliqua.</p>
```

```
    </main>
    <footer>
        &copy; 2024 My Responsive Web Page. All rights reserved.
    </footer>
</body>
</html>"""

html_file = "responsive_web_page.html"
with open(html_file, "w") as file:
    file.write(html_content)
```



Explanation:

AI tools assisted in generating the HTML structure by providing a well-organized framework with semantic elements. The responsive CSS layout was created using media queries to ensure the design adapts to different screen sizes. On desktop devices, the layout is optimized for larger screens with a horizontal navigation bar. On tablet devices, the navigation bar stacks vertically for better usability. On mobile devices, the layout is further adjusted to provide a touch-friendly experience while maintaining a clear and organized interface. The use of AI tools allowed for efficient generation of code that adheres to best practices in web design, ensuring that the web page is visually appealing and functional across all devices.